

**Log information:** The portal provides a complete log of the various activities performed by the users while accessing the portal—the number of jobs submitted by the user, the login and logout time, the job status log, job histories, various alerts regarding the job and reserving HPC resources, etc. The entire statistical details are represented through graphs and charts.

**Online tutorials and manuals:** There are online tutorials and manuals that guide users about the various functions, parameter details, environment settings, etc. of the portal.

## The Administrator Area

This is relevant to the HPC systems administrator responsible for user configuration and monitoring, job management and monitoring, queue management and monitoring, apart from issuing updates regarding the compiler and libraries. Using CHREME, the user can assume the role of an administrator with a little training in running an HPC cluster. CHREME makes this transition easier and provides the following functionalities to the administrator:

**User credential management:** CHREME's user management functionality makes it very easy to add, delete and modify users on the cluster. Through the user configuration interface, new roles can be defined for the user, various user access policies can be set up related to priority, resource allocation, assigning/un-assigning roles to users, and to disable/enable users.

**Job management/monitoring:** The portal's interface allows admins to manage and monitor all user jobs on the cluster. These include the following:

- Monitoring the status of the job and the resources allocated to it
- Controlling the job status from pre-execution state through to its completion
- Priority and queue modification for jobs starved of resources.

**Queue configuration and management:** Queue configuration allows the administrator to create classes based on various rules like the maximum number of jobs running, queued jobs, job prioritisation, nodes allocation, etc. These classes are associated with users based on how they've used the HPC system. The GUI helps in setting up various queue configurations and associating these configurations with different users. In addition to this, the GUI makes aspects of queue management like queue addition, deletion and alteration a lot easier.

**Cluster resource management and monitoring:** Through the portal's interface, cluster scheduler policies, user access policies, resource access policies, etc. can be configured.

**Compilers and libraries:** Administrators can add new compiler and library paths to the cluster environment, which provides the user with updated versions of the libraries and compilers on the HPC systems.

**Other information:** User and cluster accounting

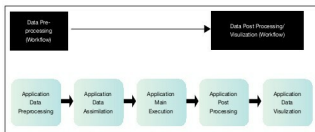


Figure 2: Application execution workflow

information can be generated by integrating accounting managers such as the 'Gold accounting manager', the periodical reports generated about overall and per-user cluster utilisation, bulletin boards, etc.

## Application-specific Interfaces

CHREME provides customised scientific application interfaces for various domains like bio-informatics, weather forecasting, computational chemistry, computational fluid dynamics, etc. These application interfaces organise, simplify and modularise the application execution process through workflows.

The portal has mechanisms to track the proper execution of the workflows, along with alerts and e-mail notification in case of any failure. "CHREME provides an easy way to integrate applications, content and provide groups of users with a single entry point to interact with their applications, data and services, all the while maintaining a uniform interface," added Kapil Mathur from the Software Development Team, HPC Solutions Group at C-DAC Pune.

Application-specific interfaces facilitate the application execution process by providing the following functionalities:

**Workflow creation:** The user can create new workflows to modularise the complete application execution process that involves data pre-processing, main application execution, post-processing and visualisation. The user can create a pipeline of several workflows, hence integrating the complete execution cycle.

**Execution:** Users can configure application-specific execution environments by defining runtime environment variables, input/output parameter values, input files, job submission parameters, etc.

**Workflow settings:** Workflow configuration like the application execution configuration can be saved, the application execution settings can be loaded and the workflow configuration settings can be modified.

**Event notification:** The workflow execution status can be notified using the portal's alerts and e-mail notifications.

## The CHREME workflow diagram

Figure 3 represents the complete flow of CHREME with

- The User Area,
- The Application Specific Interface, and